Curriculum Vitae

Dawoon Jeong

Postdoctoral Fellow
Kellogg School of Management at Northwestern University
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GitHub: https://github.com/Dawoon-Jeong0523

Professional Appointment

Northwestern University, Evanston IL, USA
Postdoctoral Fellow, Management & Organizations, Kellogg School of Management — Sep 2024 – Present
(Affiliated with the Northwestern Institute on Complex Systems)
Advisor: Hyejin Youn

Ulsan National Institute of Science and Technology (UNIST), Ulsan, Republic of Korea — Feb 2024 – Feb 2025
Lecturer, Graduate School of Interdisciplinary Management

Seoul National University, Seoul, Republic of Korea
Postdoctoral Fellow, Engineering Research Institute — Mar 2024 – Aug 2024
Advisor: Jeong-Dong Lee

EDUCATION

Seoul National University, Seoul, Republic of Korea
Ph.D. in Engineering — Mar 2020 – Feb 2024

Dissertation title: "A Study on Generalized Evolutionary Patterns of Product: Learning on Mobile Product Phylogenetic Network"
Committee: Jeong-Dong Lee, Jongsu Lee, Donghyun Oh, Junghye Lee, Yoonmo Koo

Seoul National University, Seoul, Republic of Korea
M.S. in Engineering — Mar 2018 – Feb 2020

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea
B.S. in Physics — Feb 2009 – Aug 2016

RESEARCH INTERESTS

Computational social science
Product innovation
Science of science
Network science
Economic complexity

RESEARCH METHODS

Network analysis
Machine learning
Natural language processing
Econometric

TEACHING EXPERIENCE

TEACHING

Advanced Data Mining with Artificial Intelligence (BAT51201), UNIST	Fall 2024
Text Mining (BAT54201), UNIST	Spring 2024
Analytics Programming with Application (BAT563001), UNIST	Spring 2024

PUBLICATIONS (*Corresponding Author, †Equally contributed)

Publications in Peer-Reviewed Journals

Dawoon Jeong*, Jeong-Dong Lee (2024). Where and How Does a Product Evolve? Product Innovation Pattern in Product Lineage, *Technovation, 131,* 102958.

Jeong-Dong Lee, **Dawoon Jeong***, Euy-Young Jung, Yunyoung Kim, Jiyong Kim, Youwei He, Sungjun Choi (2024). Mapping the Evolutionary Pattern of Mobile Products: A Phylogenetic Approach. *IEEE Transactions on Engineering Management. 71,* pp. 4776-4790

Hayoung Park, **Dawoon Jeong***, Jeong-Dong Lee (2024). Evolutionary mechanism for diversity dynamics in technology using a phylogenetic tree approach: directional suggestions for photovoltaic technology. *Industrial and Corporate Change.* dtae020.

Kiyoon Shin, **Dawoon Jeong***, Yoonhwan Oh. (2024). An Analysis of Global Research Space for Smart Manufacturing Based on Academic Literature. *Journal of Industrial Convergence, 22*(11), 1-11.

Dawoon Jeong, Chanlim Park*, Taewon Kang, Kiyoon Shin, Sungjun Choi, Junseok Hwang (2023). Measuring ethics level of technological topics using phylogenetic tree. *Technology Analysis & Strategic Management.*1-14.

Hoyoon Lee, **Dawoon Jeong***, Jeong-Dong Lee (2023). Drivers of institutional evolution: phylogenetic inertia and ecological pressure. *Journal of Evolutionary Economics, 33*(2).

Yeongjun Yeo, **Dawoon Jeong***, Kiyoon Shin, Jaewon Choi, You-bean Kim (2022). Analysis of evolution patterns and path dependences of the mid- to long-term strategies in the S&T field with qualitative and quantitative findings. *Journal of Korea Technology Innovation Society, 24*(5).

Reports

Anwar Aridi, Jeong-Dong Lee, Kibum Kim, Kyeyoung Shin, **Dawoon Jeong**, Wonsub Eum, Taewon Kang (2024). *Diversification through the Application of the Co-evolutionary Framework: Korea and Viet Nam*. © Washington, DC: World Bank.

IN REVIEW PROCESS (*Corresponding Author, †Equally contributed)

Dawoon Jeong, Saangkeuk Lee, Wonsub Eum*. Learning from Rivals: Diversification Paths from Countries with Similar Export Structures. *Structural Change and Economic Dynamics*. (Revise & Resubmit)

Dawoon Jeong*, Jeong-Dong Lee. Mapping Product Development Trajectories: Product Citation Network. *Scientometrics.* (Submitted)

Junghun Chae†, WooJoong Kim†, Woochul Jung†, **Dawoon Jeong†**, Roktaek Lim†, Manoj Chamlagain, Giju Jung, Juneil Jang, Jae Won Lee, Nam Kyu Kang, Kwangryul Baek, Jonghyeok Shin, Ye-Gi Lee, Hyun Gi Koh, Chanwoo Kim, Sangdo Yook, Allen Ka Loon Cheung, Yong-Su Jin, Hyejin Youn*, Pan-Jun Kim* and Cheol-Min Ghim*. Long-term innovative potential of genetic research and its suppression. *PNAS nexus.* (Submitted)

Hayoung Park, **Dawoon Jeong***, Jeong-Dong Lee. Revisiting the Routine of a Multidimensional Approach to Heterogeneous Firms: Quantification and Classification of Innovation Routines. *Journal of Evolutionary Economics.* (Revise & Resubmit)

Soohyoung Lee, **Dawoon Jeong***, Jeong-Dong Lee. Automation Will Set Occupational Mobility Free: Structural Changes in the Occupation Network, *Human and Social Sciences Communications.* (Submitted)

WORKING PAPERS

Dawoon Jeong, Seolmin Yang, James Evans, Hyejin Youn. Disruption Needs Consolidating.

Dawoon Jeong, Seolmin Yang, Sungwoo Jang, Seungryul Shin. Transitioning from science to technology on the shoulders of scientific giants: an analysis of patent-paper pair.

Dawoon Jeong*, Yeokyung Hwang, Sangnam Lee*. How many helices in the National Innovation System? Microfoundation of Triple Helix Model from R&D Actor Space.

Dawoon Jeong, James Holehouse, Jisung Yoon, Hyejin Youn. Unveiling Structure and Relationships in Legal Data: The United States Code from 1925 to 2023.

Dawoon Jeong. Moh Hosseinioun, Hyejin Youn. The Changing Role of Managers: LinkedIn Data.

Jiyong Kim, **Dawoon Jeong***, Jeong-Dong Lee. The Veiled Effort of Product Innovation: Pre-adaptation

Yeokyung Hwang, **Dawoon Jeong***. The evolution of scientific knowledge and social discourse: A multi-layer network analysis of AI technologies

Seolmin Yang, **Dawoon Jeong**, Hyejin Youn*. Technology landscape from micro to macro structure community detection

CONFERENCES

Dawoon Jeong, Jeong-Dong Lee, Jiyong Kim, Youwei He, Sungjun Choi, Joomin Lee. Evolutionary Pattern for Product Innovation: Learning on Product Phylogenetic Network

2nd International Conference on Science of Science and Innovation, Evanston, USA, Jun 26-28 (Poster)

FELLOWSHIPS & AWARDS

National Research Foundation of Korea
2023 Ph.D. Fellowship (KRW 40,000,000) Sep 2023 – Feb 2024

Korea Institute of Science & Technology Evaluation and Planning (KISTEP)
2022 KISTEP Fellowship (KRW 3,000,000) Jun 2022 - Nov 2022

Korea Institute of Science and Technology (KIST)
2021 KIST Convergence Research Fellowship (KRW 3,000,000) Jun 2021 - Oct 2021

PROFESSIONAL SERVICE

Program Committee
Atlanta Conference on Science and Innovation Policy 2025 Nov 2024 – Present

Peer review for Journals
Research Policy
Science and Public Policy

ADVISING & CO-ADVISING

Soohyoung Lee, Ph.D. Candidates, Seoul National University Sep 2024 – Present
Jiyong Kim, Ph.D. Candidates, Seoul National University Mar 2024 – Present
Yeokyung Hwang, Ph.D. Candidates, Seoul National University Sep 2024 – Present
Hayoung Park, Ph.D. Candidates, Seoul National University Sep 2022 – Present
Hoyoon Lee, Ph.D. Candidates, Seoul National University Feb 2021 – Feb 2022

WORK EXPERIENCE

World Bank
Consultant Nov 2022 - May 2023
Supported in research project "Patterns of Economic Diversification via Coevolution of Production and Technology"

Korea Broadcasting System
Consultant Feb 2023 - Mar 2023
Advised data analysis in production of documentary "*30 살 수능 길을 잃다" 시사기획 창 411 회 (KBS 23.03.28)"*

COMPUTATIONAL SKILLS

Computing Languages: Python, R
Machine Learning Libraries: Pytorch, Tensorflow
Network Visualization Software: Gephi

REFERENCES

Jeong-Dong Lee, Ph.D.
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Technology Management, Economics and Policy Program
Seoul National University, Republic of Korea
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Hyejin Youn, Ph.D.
Associate Professor
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Wonsub Eum, Ph.D.
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